

# FinFlow - Product Analytics Engineering Platform

Event-driven fintech analytics platform modeling customer behavior, transactions, retention, and product growth across savings and investment products.

## Product & Business Context

**Data Classification:** Synthetic

**Environment:** Development

**Version:** v1.0

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1. Executive Summary

FinFlow is a portfolio project simulating a production-style product analytics and analytics engineering platform for a multi-product fintech company offering digital savings and investment services across the United Kingdom and Ireland.

The project was designed to address common challenges faced by growing fintech companies — fragmented data systems, inconsistent metric definitions, inefficient reporting workflows, and limited visibility into customer behavior across the full product lifecycle.

1.1. Platform Architecture and Stack

The platform is built on a modern analytics engineering stack:

| Layer            | Technology             |
|------------------|------------------------|
| Data Generation  | Python (Numpy, Pandas) |
| Local Processing | DuckDB                 |
| Data Lake        | Google Cloud Storage   |
| Data Warehouse   | Snowflake              |
| Transformation   | dbt(Core)              |
| Orchestration    | Apache Airflow         |
| BI & Reporting   | Power BI               |

Data flows through a Medallion Architecture (Bronze → Silver → Gold), with raw events landing in S3, loading into Snowflake, and transforming into analytics-ready dimensional models via dbt. Pipelines are designed for incremental processing to support scalable daily ingestion without full-table refreshes.

Key capabilities delivered include:

- Synthetic event data generation simulating 500,000 users and over 16 million records across behavioral and transactional datasets
- End-to-end ELT pipelines with historical backfill and daily incremental ingestion
- Star schema dimensional models supporting behavioral and transactional analytics
- Automated data quality checks enforced at the Silver and Gold layers
- Product analytics use cases including funnel analysis, cohort retention, cross-product adoption, and user segmentation
- Curated Gold-layer datasets optimized for Power BI dashboards covering executive reporting, retention, product adoption, and customer growth

## **2. FinFlow - Company Overview**

Finflow is a fast-growing fintech company established in 2016, operating in the United Kingdom and Ireland. The company focuses on delivering digital savings and investment solutions tailored to retail customers.

Finflow has two core product offerings:

### **1. DigiSave**

DigiSave is a digital savings platform that enables customers to save securely with low-risk options. It offers two savings plans:

- Flexible Savings Plan (withdrawals allowed at any time)
- Fixed Savings Plan (locked savings for higher returns over a defined period)

### **2. DigiVest**

DigiVest is a digital investment platform designed for retail customers seeking low to medium-risk investment opportunities. Through DigiVest, customers can invest in:

- Government Treasury Bills
- Mutual Funds

As Finflow scales, it is generating large volumes of customer behavioral, transactional, and product interaction data across multiple systems.

## **3. Problem Statement**

Modern fintech products generate large volumes of user behavioral and transactional data across multiple channels including mobile applications, payment systems, and investment platforms. Every user action from onboarding and account funding to transfers, investments, and daily engagement produces event-level data that needs to be captured, processed, and converted into meaningful business insights.

In many early-stage and scaling fintech companies, building a reliable and scalable analytics system is still a major challenge.

Without a well-designed analytics platform, teams typically face issues such as:

- Data being spread across multiple products and systems, which makes it difficult to get a unified view of the customer
- Inconsistent definitions of key metrics like churn rate, active users, and retention across teams
- Slow query performance due to unoptimized data models and storage structures
- Heavy reliance on full refresh processes that do not scale with growing data volumes
- Limited visibility into customer behaviour across the full lifecycle and across products
- Difficulty accurately tracking funnels, retention, and cross-product adoption
- Lack of automated and reliable pipelines for daily data ingestion and transformation

This project is designed to address these challenges by building a production-style fintech product analytics and analytics engineering platform that simulates how modern digital

financial systems collect, process, model, and analyze large-scale event-driven data across multiple products.

#### **4. Project Objectives**

This project aims to design and implement a production-style fintech product analytics and analytics engineering platform that simulates how modern digital financial systems collect, process, model, and analyze behavioral and transactional data.

The platform is built to achieve the following objectives:

- Generate realistic synthetic event data across a multi-product fintech ecosystem covering savings and investments.
- Model end-to-end customer journeys to support product analytics use cases such as retention, funnels, and cross-product engagement
- Design a scalable data architecture with historical backfills and incremental daily ingestion patterns
- Implement a partition-aware data lake structure and transform raw data into analytics-ready models using dbt
- Build consistent product and growth metrics to enable reliable reporting and decision-making
- Support core analytics use cases including cohort analysis, funnel analysis, and user behavior tracking
- Simulate automated daily data pipelines using scheduled orchestration workflows
- Deliver curated datasets optimized for BI dashboards and business insights in Power BI
- Demonstrate best practices in analytics engineering including modular modeling, incremental processing, and scalable ELT design

#### **5. Customer Personas**

To support customer segmentation, product analytics, retention analysis, and behavioral modeling, FinFlow classifies users into five behavioral personas. These personas are used during synthetic data generation to simulate realistic differences in financial goals, product adoption patterns, investment behavior, and platform engagement.

While age and income influence persona assignment, personas are primarily designed to represent financial behavior and customer objectives rather than demographic characteristics alone.

##### **1. Starter Investor**

###### **Description**

Starter Investors are individuals at the beginning of their financial journey. They are typically new to investing and focused on developing healthy savings habits while learning basic wealth-building principles.

These users often begin with lower balances and smaller contributions but represent a critical growth segment because many eventually transition into long-term investors.

**Typical Characteristics**

- Limited investing experience
- Lower disposable income
- Highly mobile-first users
- Frequently make small but recurring deposits

**Financial Goals**

- Build emergency savings
- Develop consistent saving habits
- Learn investing fundamentals
- Achieve short-term financial goals

**Preferred Products**

- Flexible Savings Plans
- Fixed Savings Plans
- Entry-level Mutual Funds

**Expected Platform Behaviour**

- High onboarding completion rates
- Frequent logins
- Lower average balances
- Lower investment amounts
- Moderate retention

**Typical Demographics**

- Typically observed among users aged 18–30
- Commonly associated with annual incomes between £18,000 and £50,000

**Distribution Weight**

25%

**2. Goal-Oriented Saver****Description**

Goal-Oriented Savers primarily use FinFlow to achieve specific financial objectives. Rather than focusing on investment growth, these users prioritize disciplined saving and capital accumulation for planned future expenses.

This segment often represents highly engaged users who regularly contribute to savings products.

**Typical Characteristics**

- Consistent saving behavior
- Moderate financial literacy
- Lower risk tolerance
- Strong preference for predictable outcomes

**Financial Goals**

- Home deposit savings
- Education funding
- Travel savings

- Wedding expenses
- Emergency funds

#### **Preferred Products**

- Flexible Savings Plans
- Fixed Savings Plans

#### **Expected Platform Behaviour**

- High savings plan adoption
- Regular deposits
- Lower investment adoption
- Strong retention
- Moderate platform engagement

#### **Typical Demographics**

- Typically observed among users aged 25–55
- Commonly associated with annual incomes between £25,000 and £90,000

#### **Distribution Weight**

30%

### **3. Wealth Builder**

#### **Description**

Wealth Builders are users focused on long-term financial growth through disciplined investing. They actively balance savings and investments and are typically committed to growing their net worth over several years.

This segment represents a core target audience for many digital wealth platforms.

#### **Typical Characteristics**

- Medium to high financial literacy
- Long-term investment horizon
- Diversified product usage
- Regular investment contributions

#### **Financial Goals**

- Long-term wealth creation
- Financial independence
- Retirement preparation
- Portfolio diversification

#### **Preferred Products**

- Treasury Bills
- Mutual Funds
- Fixed Savings Plans

#### **Expected Platform Behaviour**

- High investment adoption
- Regular contributions
- Multiple active products
- Strong retention

- Moderate to high balances

#### **Typical Demographics**

- Typically observed among users aged 25–60
- Commonly associated with annual incomes between £40,000 and £150,000

#### **Distribution Weight**

25%

### **4. Active Investor**

#### **Description**

Active Investors are highly engaged users who interact frequently with the platform and regularly allocate capital across multiple investment products. They typically exhibit higher risk tolerance and a stronger focus on portfolio growth.

Although smaller in number, they often account for a disproportionate share of investment activity.

#### **Typical Characteristics**

- High financial literacy
- Frequent investment activity
- Strong product adoption
- Higher risk tolerance
- High engagement levels

#### **Financial Goals**

- Accelerated wealth growth
- Portfolio optimization
- Capital appreciation

#### **Preferred Products**

- Mutual Funds
- Treasury Bills
- Multiple concurrent investment positions

#### **Expected Platform Behaviour**

- Highest login frequency
- Highest investment frequency
- Multiple active products
- Highest average balances
- Strong retention

#### **Typical Demographics**

- Typically observed among users aged 25–65
- Commonly associated with annual incomes between £60,000 and £120,000

#### **Distribution Weight**

10%

### **5. Capital Preserver**

#### **Description**

Capital Preservers prioritize protecting existing wealth rather than maximizing investment returns. They generally prefer lower-risk financial products and value stability, predictability, and capital protection.

These users tend to maintain larger balances while exhibiting lower transaction frequency.

**Typical Characteristics**

- Conservative investment behavior
- Low risk tolerance
- Focus on wealth preservation
- Preference for stable returns

**Financial Goals**

- Protect accumulated wealth
- Generate predictable returns
- Preserve purchasing power
- Maintain financial security

**Preferred Products**

- Treasury Bills
- Fixed Savings Plans

**Expected Platform Behaviour**

- Lower login frequency
- Larger average balances
- Longer holding periods
- Lower product experimentation
- Strong retention

**Typical Demographics**

- Typically observed among users aged 45+
- Commonly associated with annual incomes between £70,000 and £150,000

**Distribution Weight**

10%

**5.1. Persona Assignment Logic**

Customer personas are assigned during synthetic data generation based on a weighted combination of:

- Age
- Reported annual income
- Savings propensity
- Investment propensity
- Product preferences

**5.2. Persona Distribution**

| Persona          | Weight |
|------------------|--------|
| Starter Investor | 25%    |



|                     |     |
|---------------------|-----|
| Goal-Oriented Saver | 30% |
| Wealth Builder      | 25% |
| Active Investor     | 10% |
| Capital Preserver   | 10% |

### 5.3. Product Adoption Expectations by Persona

| Persona             | Savings Adoption | Investment Adoption | Retention | Engagement |
|---------------------|------------------|---------------------|-----------|------------|
| Starter Investor    | High             | Low                 | Medium    | High       |
| Goal-Oriented Saver | Very High        | Low                 | High      | Medium     |
| Wealth Builder      | High             | High                | High      | High       |
| Active Investor     | Medium           | Very High           | High      | Very High  |
| Capital Preserver   | High             | Medium              | High      | Low        |

### 5.4. Customer Acquisition Channels

| Channel          | Definition                                                                      | Weight |
|------------------|---------------------------------------------------------------------------------|--------|
| Organic Search   | Users acquired through unpaid search engine traffic                             | 25%    |
| Paid Social      | Users acquired through paid advertising campaigns on social media platforms     | 20%    |
| Referral Program | Users acquired through customer referral initiatives                            | 20%    |
| Direct Traffic   | Users who directly access the platform through known URLs or bookmarks          | 15%    |
| Paid Search      | Users acquired through search engine advertising campaigns                      | 10%    |
| Partnerships     | Users acquired through strategic partnerships, affiliates, or employer programs | 10%    |

## 5.5. Customer Acquisition Channel by Persona

| Persona             | Organic Search | Paid Social | Referral Program | Direct Traffic | Paid Search | Partnerships |
|---------------------|----------------|-------------|------------------|----------------|-------------|--------------|
| Starter Investor    | 15%            | 40%         | 20%              | 5%             | 10%         | 10%          |
| Goal-Oriented Saver | 30%            | 15%         | 20%              | 15%            | 10%         | 10%          |
| Wealth Builder      | 25%            | 10%         | 20%              | 20%            | 10%         | 15%          |
| Active Investor     | 20%            | 10%         | 15%              | 30%            | 15%         | 10%          |
| Capital Preserver   | 30%            | 5%          | 25%              | 25%            | 5%          | 10%          |

## 6. Event Tracking Philosophy

FinFlow follows an event-driven analytics model where every meaningful customer interaction is captured as a timestamped event.

These events serve as the foundation for:

- Product analytics
- Funnel analysis
- Retention analysis
- Customer journey analysis
- Product adoption reporting

### Examples

- signup\_completed
- app\_login
- kyc\_completed
- wallet\_funded
- savings\_plan\_created
- investment\_plan\_created
- investment\_vests
- early\_withdrawal\_initiated

## 7. Business Requirements

### 7.1. Product Analytics Requirements

#### 1) Onboarding & Activation Analytics

##### Requirements:

- Monitor onboarding completion
- Detect onboarding drop-offs
- Measure activation rates

- Improve time-to-value

#### **KPIs**

- KYC completion rate
- Activation Rate
- D1 Retention
- Time to first Transaction

#### **Events**

- signup\_completed
- kyc\_completed
- wallet\_funded
- login

- 2) **Engagement Analytics:** This helps the business understand how actively users use the product.

#### **Requirements**

- Track product and plan adoption and usage
- Measure session frequency
- Analyze user journeys

#### **KPIs**

- DAU/MAU
- Session duration
- Product / Plan adoption rate

#### **Events**

- app\_login
- savings\_plan\_created
- investment\_plan\_created

- 3) **Retention & Churn Analysis:** The business wants to keep users active long-term.

#### **Requirements**

- Measure cohort retention
- Identify churn signals
- Detect disengaged users
- Understand why users leave

#### **KPIs**

- D1/D7/D30 retention
- Churn rate
- Repeat transaction rate
- Cohort retention curves

- 4) **Conversion Funnel Analytics:** The business wants to optimize user conversion paths.

#### **Requirements**

- Track funnel progression
- Detect abandonment points
- Improve conversion rates

#### **Funnel Flow**

- Signup → **KYC Completed** → **Wallet Funded** → **First Product Adoption** (Savings Plan OR Investment Product) → **Repeat Product Usage**

#### KPIs

- Funnel completion rate
- Drop-off rate
- Conversion rate

- 5) **Product Adoption Analytics:** The business wants to understand how the product is received and interacted with by the users

#### Requirements

- Measure savings product adoption
- Measure investment product adoption
- Track cross-product usage
- Understand product engagement

#### KPIs

- Savings Plan Adoption Rate
- Investment Adoption Rate
- Cross-Product Adoption Rate
- Average Products per User

- 6) **Customer Segmentation:** The business wants to analyze different user groups separately.

#### Requirements

- Segment users by behavior
- Compare cohorts
- Personalize experiences

#### Common Segments

- New vs returning users
- Free vs premium users
- High-value investors
- Dormant users

- 7) **Operational Monitoring:** The business wants visibility into product/system health.

#### Requirements

- Monitor failed transactions
- Detect latency issues
- Identify broken flows

#### KPIs

- Payment success rate
- API failure rate
- App crash rate

- 8) **Executive Reporting & Decision Support:** Leadership summarised business insights

## **Requirements**

- Executive dashboards
- KPI reporting
- Trend analysis
- Growth dashboard
- Retention dashboard
- Product adoption dashboard
- Customer growth dashboard
- Funnel dashboard

## **7.2. Data Engineering Requirements**

### **1. Data Ingestion Layer**

#### **Sources**

The platform shall ingest data from synthetic datasets representing:

- User profiles
- User events
- Wallet transactions
- Investment Positions (Savings, Treasury Bills, and Mutual Funds)
- Investment transactions
- Product reference data

#### **Requirements**

- Support historical backfill of three years of data.
- Support daily incremental data generation.
- Store source files in Parquet format.
- Maintain source system lineage.
- Enable repeatable and idempotent loads.

### **2. Data Lake Requirements**

#### **Storage Layer**

The platform shall utilize Google Cloud Storage as the centralized data lake.

#### **Requirements**

- Store raw source files without modification.
- Partition event and transaction datasets by date.
- Maintain historical snapshots.
- Support scalable storage of high-volume event data.
- Enable efficient downstream processing.

### **3. Data Warehouse Requirements**

#### **Architecture**

The platform shall implement a Medallion Architecture consisting of:

#### **Bronze Layer**

- Raw source data.
- 1:1 representation of source files.
- Minimal transformation.

#### **Silver Layer**

- Data cleansing.

- Standardization.
- Deduplication.
- Business rule enforcement.
- Data quality validation.

#### **Gold Layer**

- Analytics-ready data marts.
- Star schema modeling.
- Product analytics datasets.
- BI-optimized structures.

### **4. Data Modelling Requirements**

The platform shall support both behavioral and transactional analytics.

#### **Core Dimensions**

- dim\_date
- dim\_user
- dim\_wallet
- dim\_product
- dim\_plan
- dim\_event\_type

#### **Core Facts**

- fact\_user\_event
- fact\_investment\_position
- fact\_transaction

#### **Requirements**

- Use surrogate keys for all dimensions.
- Enforce referential integrity.
- Support historical trend analysis.
- Optimize for analytical query performance.

### **5. Data Quality Requirements**

The platform shall enforce automated data quality controls.

#### **Requirements**

- Primary key uniqueness validation.
- Foreign key relationship validation.
- Null value checks.
- Date validation.
- Duplicate record detection.
- Business rule validation.

#### **Example Rules**

- Users must be at least 18 years old.
- Investment amount must be greater than zero.
- The amount invested must be less than wallet balance at event time.
- Event timestamps cannot occur in the future.
- KYC completion must occur after signup.

- Event sequence needs to follow the project event flow

## 6. Incremental Processing Requirements

The platform shall support scalable incremental processing.

### Requirements

- Process only newly generated records.
- Avoid full-table refreshes where possible.
- Support partition pruning.
- Support incremental dbt models.

## 7. Orchestration Requirements

The platform shall automate daily pipeline execution.

### Requirements

- Trigger daily synthetic data generation.
- Execute ingestion workflows.
- Run dbt transformations.
- Execute data quality tests.
- Log pipeline execution status.

## 8. Monitoring & Observability Requirements

The platform shall provide visibility into pipeline health.

### Requirements

- Pipeline execution monitoring.
- Data freshness monitoring.
- Row count reconciliation.
- Failure alerting.
- Transformation auditability.

## 9. Business Intelligence Requirements

The platform shall provide curated datasets optimized for Power BI.

### Requirements

- Funnel Analysis
- Cohort Analysis
- Retention Analysis
- User Segmentation
- Product Adoption Analysis
- Executive KPI Reporting

## 8. Expected Dataset Scale

| Entity                           | Approximate Volume |
|----------------------------------|--------------------|
| Users                            | 500,000            |
| User Events                      | 12,000,000         |
| Investment & Wallet Transactions | 5,000,000          |

|                      |           |
|----------------------|-----------|
| Investment Positions | 1,100,000 |
|----------------------|-----------|

## 9. Business Assumptions

- Minimum user age is 18 years.
- KYC is required before any investment activity.
- Users must fund a wallet before creating a savings or investment position.
- Users may hold multiple products simultaneously.
- Savings plans and investment products are independent.
- All data is synthetic and generated for analytical purposes.
- Product adoption behavior varies by customer persona.
- Cross-product adoption is expected to increase over the customer lifecycle.